ENGINEERING, PROCUREMENT CONSTRUCTION & COMMISSIONING

[Internal]

[UTTHUNGA]
[Indiqube, South Summit, South End Rd, Vijayarangam Layout, Bengaluru, Karnataka 560004]

Contents

1 Purpose 3
2 Scope 3
3 Abbreviations 3
4 Terms and Definitions 4
5 Responsibilities 4
6 RACI 5
7 SIPOC 6
8 Overall Process Flow Chart 8
9 Process Description 10
9.1 Pre-bid Stage 10
9.2 Project Planning 10
9.3 Project Execution (Construction), Monitoring & Control 12
10 Project Handover & Closure 16
11 Information Security requirements 17
12 References 18
13 Metrics 18
14 Annexures 19
15 Forms 19

CHANGE HISTORY

This section describes the details of changes that have resulted in the current document.

SL. No.	Document Version No.	Change Description	Reason For change	Date Prepared By	Date Reviewed By	Date Approved By
01	0.1	Draft	Implementation of the IMS	Abithith 19-Feb-24		
02	Issue B V1.0	Revised the Document to suit the requirements for implementation of the IMS			Arun Prasad M 11-Mar-23	Sanjeeva G Byakod 11.04.2024 Sanmangala MS 12-Apr-24
03	Issue B V1.1	Document formatting done as per document guidelines		Obul Prasad Reddy 22-Dec-24	Anitha N 23-Dec-24	Santosh V 23-Dec-24
04	Issue B V1.2	Metrics and Forms updated	Updated the Metrics and new forms for implementation	Aniket Anant Prab 18-Mar-25	Arun Prasad M 18-Mar-25	Santosh V 28-Mar-2025
05	Issue C Version 1.0	Review of document	New version update of IMS	Anitha N 01-Aug-25	Santosh V 08-Aug-25	Sanmangala MS 14-Aug-25
06	Issue C Version 1.1	Revised section 9.3 to include customer escalation & complaint process	To include Customer escalation and complaint process	Aniket Parab 10-Oct-25	Santosh V 10-Oct-25	Sanmangala M S 10-Oct-25
07	Issue C Version 1.2	Revised section 9.2 #m to include BBU requirement and section 15 to add details of the new templates	To include BBU requirement and details of the new released templates and forms	Aniket Parab 11-Nov-25	Santosh V 13-Nov-25	Sanmangala M S 14-Nov-25

1 Purpose

To define a standard procedure for Engineering, Procurement Construction, and commissioning activities at Utthunga.

2 Scope

The Scope of this SOP is to lay down the requirements for a managing Engineering Procurement Construction and Commissioning Projects with Utthunga.

3 Abbreviations

a. AGM : Assistant General Manager
b. BOM : Bill of Material
c. BOQ : Bill of Quantities
d. BU : Business Unit
e. DLP : Defect Liability Period
f. DPR : Detailed Project Report
g. DC : Document Control
h. DWPR : Daily Work Progress Reports
i. EOI : Expression of Interest
j. EPC : Engineering Procurement & Construction
k. FEED : Front-End Engineering and Design
l. GM : General Manager
m. GST : Goods and Service Tax
n. HSE : Health Safety & Environment
o. ITP : Inspection & Testing Plans
p. MoM : Minutes of Meeting
q. MWPR : Monthly Work Progress Report
r. NCR : Non-Conformance Report
s. PM : Project Manager
t. PO : Purchase Order
u. PPE : Personal Protective Equipment
v. QM : Quality Manager
w. QAP : Quality Assurance Plan
x. RFI : Request for Information
y. ROW : Right of Way
z. WWPR : Weekly Work Progress Report
aa. GRN : Goods Receipt note
bb. IR : Inspection Report
cc. BBU : Billing Breakup Unit

4 Terms and Definitions

a. In this document the terms Customer(s) & Employer(s) are used interchangeably, all these usages are for denoting Customer(s), for whom or on whose behalf Utthunga provides services and, or products.

b. Similarly, Vendor(s), Subcontractor(s) & Supplier(s) are used interchangeably, and all these usages are for denoting Vendor(s), who provides services & products to Utthunga or providing them to Customer(s) on behalf of Utthunga.

c. Concept Development: - Understanding the priority and objective requirements of the customer for the facility under development and formulating a conceptual design. Utthunga will design a facility or retrofit based on the capacity, inputs, outputs, throughput, and other specifications as per the customer's needs.

d. Feasibility Studies: - Once concepts are being identified, Utthunga will conduct feasibility studies to assist the Customer in evaluating options. In this stage, the budget and design are both refined. Better estimates and defined scope, enables the Customer to determine which concept is most suitable.

e. Front-End Engineering and Design (FEED): - Embracing the Customer preferences and industry standards Utthunga will move forward with FEED. Preliminary design, estimations, analysis etcetera will begin. The documentation created during FEED will also assist during the bidding as well as detailed engineering phase. FEED will also produce a project cost estimate that is suitable for project financing decisions.

f. Detailed Design & Engineering: - With all the necessary data obtained Utthunga will develop a detailed engineering package in which all the necessary documents, details, drawings, managing tools etcetera are available. These documents shall be the baseline for the entire project planning, construction phases as well as for operation of the facility.

g. Engineering, Procurement & Construction: - Once construction begins with approval from Customer as well as from all authorities concerned Utthunga will manage all the resources required for the project including human resource as per the contract. The pre-construction planning will guide construction phase. Project Manager(s) will proactively troubleshoot any issues on the job site and ensure that the facility is completed with quality craftsmanship. Utthunga will also be responsible for managing the project timeline and budget for the contracted scope.

h. Commissioning & Operations: - Commissioning is a sophisticated process and Utthunga with its proficient workmanship ensure that the facility up and running without any delays or constrains.

5 Responsibilities

a. Business Unit Head shall be responsible of overall operation of EPC business unit.

b. Quality Head shall be responsible for Implementation of QMS system and projects are delivered as per the system and Customer requirements.

c. Project Manager involves in planning, assist in procurement and executing the project as scheduled.

d. Quality Manager is responsible for the ensuring the compliance as required by customer and implemented QMS requirements.

e. Procurement Manager shall be responsible of arranging the resources other than manpower and liaise with vendor for successful completion of project.

f. Planning Engineer involves in prepare, update & execute the project in liaison with all support functions.

g. Quality Engineer shall be responsible for inspecting, verification of incoming materials and as per approved inspection and test plan/project quality plan.

h. Sub vendor shall be responsible for supplying, providing services as stated by UTT purchase order and project requirements.

6 RACI

Sl. No.	Process	Responsible	Accountable	Consulted	Informed
1	Lead & Opportunity Management	Sales Manager	BU Head	Finance Manager Design Manager Project Manager Quality Manager	CEO/MD
2	Inhouse Project Team Selection	Project Manager Quality Manager Design Manager	Project Manager HR Manager	BU Head Quality Head	CEO/MD
3	Risk Identification & Assessment	Project Engineer Quality Engineer Safety Officer Procurement Engineer	Project Manager Quality Manager Safety officer Procurement Manager	BU Head Quality Head	CEO/MD
4	Preparation of Plan, Schedule, tracking tools etc.	Project Engineer	Projects Manager	Design Manager Procurement Manager Quality Manager	BU Head
5	Project Execution Design Process Quality Process Procurement process HSE	Design Engineer Project Engineer Quality Engineer Procurement Engineer Safety officer	Design Manager Project Manager Quality Manager Procurement Manager Safety Officer	Delivery/Operations Head Quality Head BU Head	CEO/MD

6	Change order/ Extra item tracking	Project Engineer	Project Manager	BU Head	CEO/MD
7	Project Health Monitoring, Review & Reporting	Project Manager	Quality Manager Design Manager Procurement Manager Sub vendor/Suppliers HSE Officer	BU Head Quality Head	CEO/MD Finance Manager
8	Pre-commissioning, Testing & commissioning	Commissioning Engineer, Quality Engineer	Project Manger Quality Manager HSE officer	BU Head Quality Head	CEO/MD
9	Providing Training & Handing over Documents	Project Manager Quality Manager	Project Manager Quality Manager	BU Head Quality Head	CEO/MD Customer Finance Manager
10	Demobilizing Resources	Project Manger HSE officer	Project Manager	BU Head	CEO/MD
11	Information for Invoicing	Project Manager Finance Manager	EPCC Delivery Head Finance Manager	Project Manager Finance Manager	CEO/MD

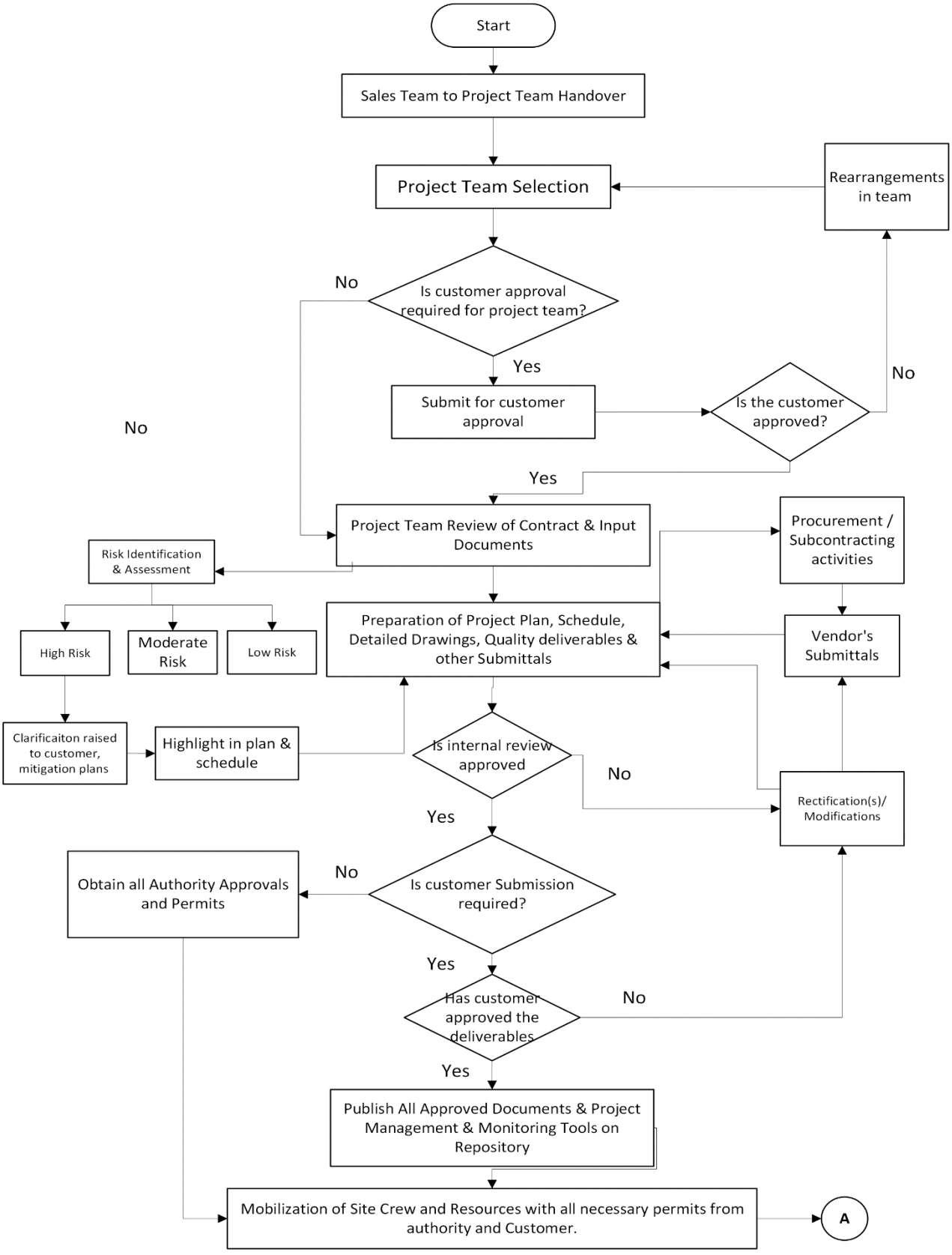
7 SIPOC

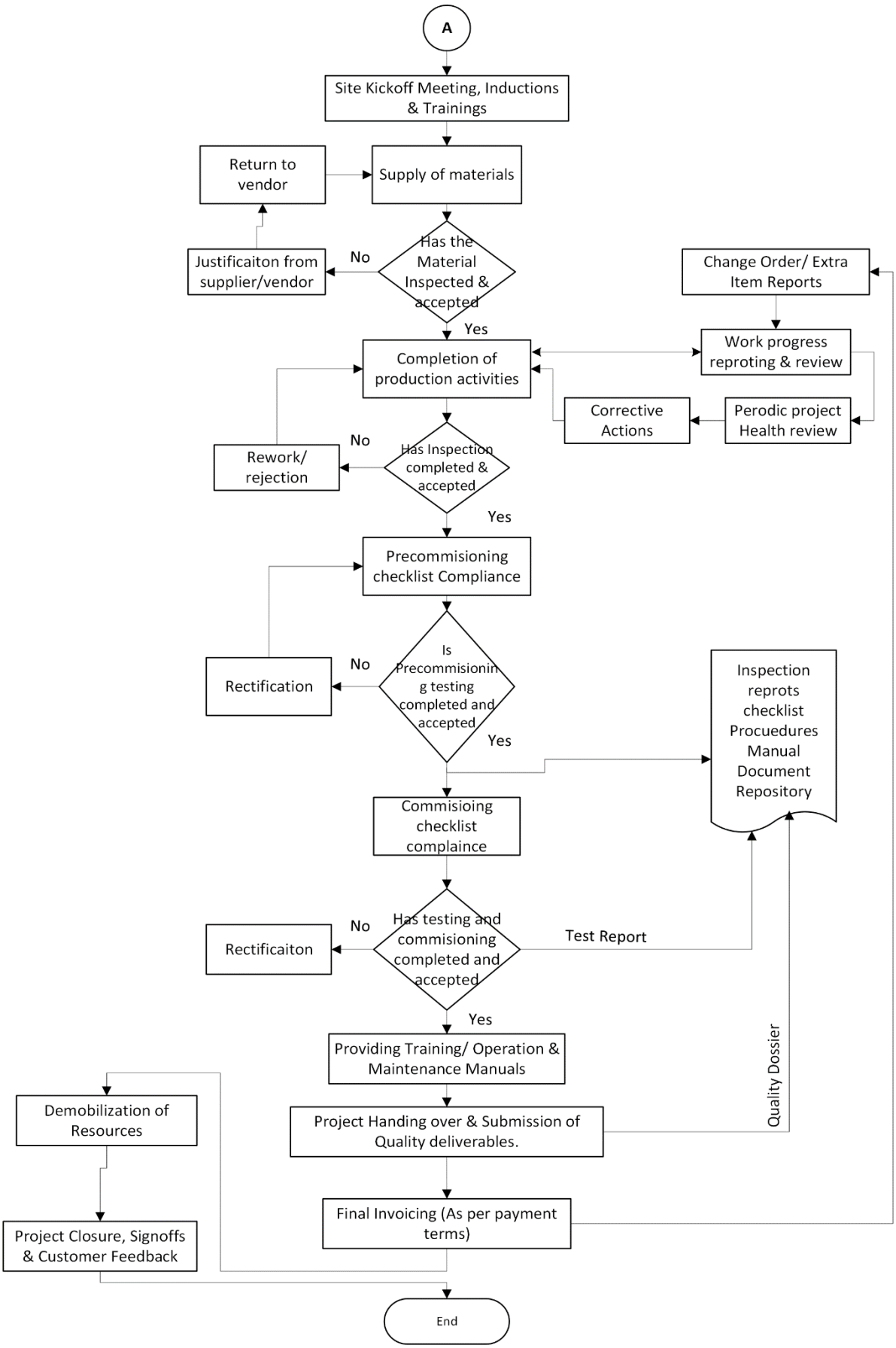
Supplier	Input	Process	Output	Customer
Sales Team	Contract Details	Sales to Delivery handover	Sales to Delivery handover Form	Project Manager
Sales Team	Contract Details, Sales to Delivery Handover form	Project Planning	Approved Project Plan	Project Manager, Customer
Project Manager	Approved Project Plan	Allocation of resources	Project team member list	Project manager, HR
Project manager, HR	PP, Project team member list	Project Execution	Project delivery documents, Test reports, commissioning	Project Manager, Customer

			completion certificate, MRB	
Project Manager	Project delivery documents, Completion certificate	Project Closure	Project Closure checklist, Archival of project related items, Resources releasing, Contract closure, CSAT	Customer

8 Overall Process Flow Chart

Project Phase: Planning, Execution, Monitoring & Control; Handover & Closure.




A
Site Kickoff Meeting, Inductions & Trainings
Return to vendor
Supply of materials
Justificaiton from supplier/vendor
No
Has the Material Inspected & accepted
Yes
Change Order/ Extra Item Reports
Work progress reproting & review
Completion of production activities
Corrective Actions
Perodic project Health review
Rework/ rejection
No
Has Inspection completed & accepted
Yes
Precommisioning checklist Compliance
Rectification
No
Is Precommisioning testing completed and accepted
Yes
Inspection reprots checklist Procuedures Manual Document Repository
Commisioing checklist complaince
Rectificaiton
No
Has testing and commisioning completed and accepted
Test Report
Yes
Providing Training/ Operation & Maintenance Manuals
Quality Dossier
Demobilization of Resources
Project Handing over & Submission of Quality deliverables.
Final Invoicing (As per payment terms)
Project Closure, Signoffs & Customer Feedback
End

9 Process Description

9.1 Pre-bid Stage

a. The standard operating procedure lead management Latest version provides details of how the enquiry from Customers are processed.

b. Finalizing the orders based on the enquiry form Customers is addressed in Opportunity Management standard operating procedure.

9.2 Project Planning

a. Utthunga shall formulate a comprehensive plan for the entire project during the planning stage, so that the systems are set up to manage the project once its execution commences. A well-defined Project Plan & Process serves as a constant reminder of the tasks that need to be performed to increase the likelihood of achieving a successful project outcome.

b. Selection of Project Team shall be finalized based on project requirement including the selection of Project Manager, Design Engineers, Project Engineers, Field Supervisors, and other key team members as required. Decisions regarding in-house deliveries and outsourcing shall be finalized; selection of in-house team shall be approved by BU Head or person designated by BU Head.

c. Project team shall review contract for unfavorable or high-risk clauses and should also review specifications, plan, schedule of the customer and should create lists of issues that needed to be resolved and the same shall be communicated and discussed with customer and resolved.

d. Detailed Project Schedule and Plan with major milestones, resources and risks identified shall be prepared and its conformity to various lead times, delivery obligations, resource availabilities, budget, BOQ line items, risk factors, quality aspects shall be verified and ensured.

e. Risk assessment shall be conducted as per project risk and opportunity management templates in terms of schedule, resource utilization, deployment, planning etc. The risk shall be identified into low, medium, and high; mitigation plans for the identified risk shall be derived, in case of elevated risk the same shall be informed to Customer and If Deviations required in technical and product related a change request will be raised to customer approval for making alternative arrangements.

f. Design Process shall be followed for detailed Engineering drawings, Shop Drawings, Panel Drawings, Internal General Arrangement Drawings, General Arrangement Drawings, Line Diagrams, Control Wirings, Typical Detail Drawings, Layout Plans, Control Topologies, Sequence of Operation Matrix, Bill of Quantities (BOQs) etcetera shall be created using the conceptual design inputs or input drawings from the Customer, Consultant or anyone engaged by Customer and ensure that the project design and specifications are conforming to all the relevant standards, specifications, governing regulations.

g. All the drawings, blueprints, distribution maps, layouts, schematic diagrams, equipment specifications, design reports, drawing transmittal records shall be reviewed & approved by the Design Manager or the person designated by BU Head at various stages.

h. Project Manager Shall be responsible for preparation of project schedule, planning, registers, and logbooks for change request. Design Manager shall be responsible for preparation review and approval of design, drawings, schematic diagrams and BOQ, BOM, Quality Engineer shall prepare the QAPS, ITP and Project manager Shall be responsible for approval of project quality assurance plan, Inspection and test plan and method statements and checklist used for Monitoring and Auditing Tools.

i. The quality related documents such as project quality plan, Inspection related documents, technical specifications, manuals such as commissioning, operation shall be reviewed and approved by Project manager prior to submission for Customer approval.

j. The Project Manager/ Document Controller/Project Control Engineer as the full control of submissions and correspondences. This includes submission and tracking of Subcontractor(s)/ Vendor(s) submittals after verification.

k. Vendor(s)/ Subcontractor(s) Management shall be as per the project requirements Sub vendors and supplier shall be engaged. The selection and engaging of Sub vendor and supplier shall be as per procurement standard operating procedure. The engagement of Subcontractor(s)/ Vendor(s) shall be duly communicated with the Customer & all the concerned as per the requirements in Terms and Conditions of the Contract/purchase order.

l. The project Manager shall expedite for the statutory approvals whichever are in the scope of Utthunga and if the approvals are in part or wholly on Customer's scope, Utthunga shall ensure that the same is obtained including for ROWs and authorization to commence work.

m. Obtain the BBU approval from customer to be inline with the invoice/payment schedule.

n. Once the approvals are obtained from the Customer/ Consultant for the submittals, meetings shall be organized among various stake holders, including among project team members (including site execution team), with Subcontractor(s)/ Vendor(s)/ Supplier(s), with Customer's representatives, Consultants etc. Actions to be initiated by all the stakeholders according to the Project Plan and Schedules are once again reviewed and consensus is obtained before the actual commencement of the execution of project. The site activities shall not be commenced until formal approval received from customer on the submitted relevant project documents.

o. Throughout the project lifecycle all the documents except confidential and classified ones shall be kept in the centralized repository of the Company in the specified folder structure so that all the stakeholders can readily access and view them, Delivery /Operations Head shall ensure that all the Team Members to follow this.

9.3 Project Execution (Construction), Monitoring & Control

a. Before mobilizing or allocating resources, the project manager shall ensure that all the necessary approvals for starting the project has been obtained and that there is necessary site clearance for mobilization as well as for the commencement of work.

b. The mobilization activity shall be included in the project schedule/plan and any deviation from the actual approved schedule shall be duly communicated as per the project communication matrix and recorded.

c. Safety Officer shall visit the site before allocating human resource to the project site. Safety officer shall carry out project specific hazards identification and risk assessment in consultation Customer's/ Main Contractors.

d. Further, the HSE representative shall ensure that there are sufficient facilities like Toilets, Electricity, Drinking Water, Dining Facilities, Rest Areas, First Aid and Emergency Response Facilities, Communication Facilities, Site Office, or provision for providing such facilities available in the site for mobilizing human resource.

e. Safety officer shall formulate a project specific comprehensive HSE plan in accordance with the Company's HSE policies and project quality assurance plan and shall ensure that safety procedures and work practices are followed on each part of the project right from the beginning till handover and further.

f. After the receipt of necessary approvals, permits and site clearance form customer, project manager initiates the mobilization of crew, tools, and machinery to the site. The mobilization of crew and machinery is addressed in the project plan, if any deviation in the plan shall be escalated to customer and Delivery/Operations Head or person designated by BU Head to facilitate smooth flow of project execution.

g. All the project activities shall be carried out as per the Project Schedule and Plan, any deviations or changes from the approved plan shall be duly recorded and communicated to the concerned. Project Manager shall ensure that all the deliverables are completed within the scheduled date and time.

h. Project Manager, Project Engineers, Site In charge, Supervisor, sub vendors etcetera shall ensure that all the works executed are as per the approved drawings, specifications, governing standards, best and safe work practices, this also applies to the works executed by Subcontractor(s).

i. The Deviations are raised when there is a divergence in BOM/BOQ, drawing requirements and there will not be any cost implications. Change requests are raised when there is change in requirement or alternative solution implemented due to any changes irrespective of whether it is site condition requirements, Customer/ Consultant requirements or due to any other reason shall be recorded on Change Order/ Deviation Statements and approvals shall be obtained prior to actual site

execution. This shall also be followed for any extra work needed to be executed than that mentioned in BOQ/ BOM. In both cases prior approval from Company's BU Head & Finance Manager shall be obtained before submitting Change Order/ Deviation/ Extra Item approval requests to Customer/ Consultant.

j. Job Site & Coordination Meetings shall be conducted for effective and frequent communication among project stakeholders the Project Manager, Site Supervisor, should attend jobsite coordination meetings to work out any conflicts, including that with other agencies before the conflicts/ constrains delays the work. The coordination at the beginning of the project is specifically important and it shall be conducted as a Site Kickoff Meeting with a well-defined agenda so that important items are not missed out.

k. All-important communications shall be well timed, through designated channels only and addressed to all the relevant parties. Communication happens at every level, all the formal communications shall be properly recorded and updated, action items shall be labelled important.

l. Any communication received from the customer — through email, meeting minutes, site reports, or verbal escalation — that highlights a deviation, concern, or dissatisfaction related to project deliverables, performance, or services shall be evaluated as either a Project Issue or a Customer Complaint.

- **Project Issue:** A Project Issue refers to an operational or technical concern raised by the customer that:
  - Arises during project execution but does not indicate customer dissatisfaction or breach of requirements.
  - Is manageable within the project team's scope, with immediate corrective action possible at site or through coordination.
  - Does not result in cost, schedule, or quality impact beyond acceptable project tolerances.
  - Typically includes clarifications, document updates, minor non-conformities, or field-level corrections.

  Project Issues will be logged in the Issue Register and tracked to closure by the respective Project Manager or Site Lead.

- **Customer Complaint:** A Customer Complaint refers to a communication from the customer that:
  - Expresses dissatisfaction with the product, service, or project performance.
  - Indicates a failure to meet contractual, quality, or performance requirements.
  - Has potential or actual impact on cost, delivery, compliance, or reputation.
  - Is escalated beyond the project manager/site lead or involves repeated occurrences of similar issues.

Customer Complaints shall be logged in the Customer Complaint Register and shall require Root Cause Analysis report as defined in the SOP - Customer Complaint and Escalation UT/PS/CCEP/SOP.

m. Approval for extension of Completion period, shall be obtained well in advance to avoid any unnecessary implications and the communications regarding the same shall only be through proper formal channel. If Government Departments/ Utilities/ or similar highly regulated agencies are involved, then the extensions/ change orders/ extra works/ price revisions etcetera shall be in the form of approved valid documents from the concerned authority rather than electronic communication. Any verbal communication has been initiated by either party shall be recorded in MOM and the same shall be circulated as per the communication matrix. If any compensation provision/ other implications are included as part of the extension it shall be notified to the CEO, Finance Manager & Delivery/ Operations Head by project manager.

n. The project reporting and documentation procedures shall be followed right from the beginning. The Project Team shall use the approved Project Reporting Tools to the extent possible, if any changes or additions are required it shall be made only after obtaining approval from the Delivery /Operations Head. However, Project Team shall be empowered to prepare tools and track projects based on wide range of criterion for internal analysis purpose.

o. All the data collected throughout the project lifecycle shall be recorded in the Project Control and Monitoring data sheets in comprehensive form and Project Manager, Project control Engineer should follow the specified folder structure for keeping project documents.

p. Project management and reporting including Work Progress Reports like Daily Work Progress Reports (DWPRs), Weekly Work Progress Report (WWPRs) and Monthly Work Progress Report (MWPRs), Project Status Reports, Minutes of meeting's, clarifications, Break-up Schedules, Changes Orders, Short-term Plans, Specifications, Transmittal Records, Submittal updates, Photographs, and correspondence shall be updated on the document repository of the Projects. The Project Manager/Document Controller/project control engineer shall update the documents and files properly and timely as per the Document Checklist.

q. All the project departments including Execution, Design& Planning, Procurement, Accounts shall perform their duties with due diligence and in coordination for the well timed and successful delivery of the project. This includes supply of materials as per the supply schedules, execution of site works as per work schedule, preparation of revision and addition drawings, submission of various documents/ certificates required by the Customer/ Consultant from time to time,

r. This also includes the proper communication of the project activities and progresses to the Customer/ Consultant, assisting the concerned in inspections, raising work or material inspection requests with the Consultants or third parties, submitting relevant documents and test certificates in various stages of the project, arranging/ conducting/ inspecting/ co-inspecting/ various pre-dispatch tests like Type Test,

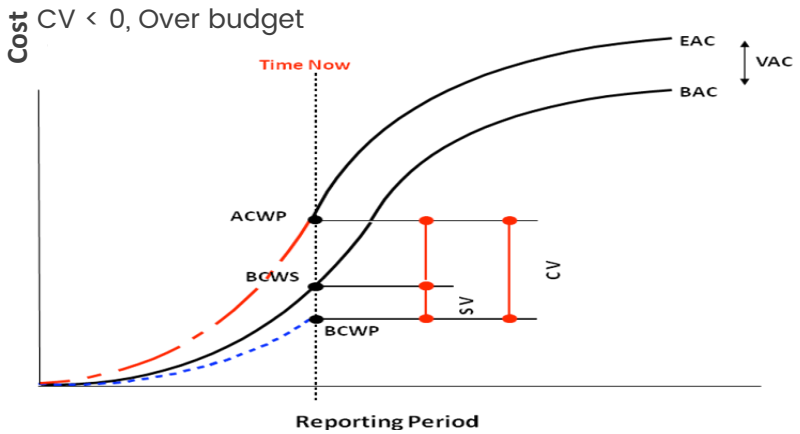
Factory Acceptance Tests etcetera, Receiving Material Inspections and raising inspections for the same etcetera and obtaining Manufacturing & Dispatch clearances where ever applicable as per project requirement.

s. Billing and Cost Control is established in the Project Initiation and Planning stage. Billing involves the submission of invoices for work and materials that has been delivered. Invoices shall be raised as per the contract terms and with utmost accuracy as well as well-timed.

t. Change order costs should be tracked separately from the original scope of work to ensure that it is never missed out and the cost growth can be properly accounted.

u. The Delivery/Operations Head shall monitor the Project Health throughout the entire lifecycle of the project and review meetings shall be conducted regularly and any corrective measures shall be suggested or implemented. The Delivery/Operations Head shall also update budget and delivery timelines based on the factors that are influencing project delivery like Change Orders, delays, constrains etc. The Actual Project Costs, Budgeted Cost of Work Scheduled and Earned Value etcetera can be compared to track and forecast the overall health of the project.

- Budgeted Cost for Work Performed (BCWP).
- Budgeted Cost for Work Scheduled –BCWS/ACWP
- Actual Cost of Work Performed -ACWP
- Schedule Variance (SV) = BCWP –BCWS
  - SV > 0, Ahead of schedule
  - SV < 0, Behind schedule
- Cost Variance (CV) = BCWP –ACWP
  - CV > 0, Under budget
  - CV < 0, Over budget



v. The Project Manager shall ensure that all upon completion of each milestone activities including inspection and standalone testing and documentation shall raise request with Customer/Consultant for starting standalone testing of the systems.

w. In the work based Standard Testing and Commissioning Report (Draft) the Commissioning Engineer Shall furnish the details of the outcomes of standalone Testing activity and shall prepare a report highlighting any deviation from the

expected outcome and the same shall be shared to the Project Manager and the PM shall take necessary actions with proper RCA to rectify the defects on works executed/ materials/ equipment.

x. After completion of all the major works in the Project and as per schedule the Project Manager shall request for Testing and Commissioning of the systems. Before which the Project Manager shall ensure that all the related works are completed including that of other agencies associated with the system and all preceding inspections including the work inspections as well as the pre-commissioning inspections are completed and approved, all NCRs are closed, and all previous test fails are addressed and rectified.

y. The Project Manager shall ensure the availability of Qualified Commissioning Engineer(s) and Technician(s) during the pre-commissioning inspection as well as Testing and Commissioning activity, and the activity shall only be carried out after receiving approvals from all the departments and stakeholders including Safety officer Customer, Consultant etcetera, Testing and Commissioning shall be carried out only in the presence of the concerned parties.

z. Once Testing and Commissioning is successfully carried out work based Standard Testing and Commissioning Report shall be prepared by the Commissioning Engineer and handed over to the Project Manager. The site in charge shall submit the Testing and Commissioning Report to the Project Manager and Quality Manager for approval and the approved document shall be submitted to the Customer/ Consultant/ Concerned Authority for approval by project manager.

aa. Project Manager shall ensure that all the works including minor items as per the Project Plan and Schedule, BOQ, Change Orders, Extra-deviation Item requirements are executed in complete and shall raise request for inspections and obtain approvals for the works/ supplies whose approvals are pending and for any missed-out items.

10 Project Handover & Closure

a. Project Manager shall ensure that all the deliverables are complete as per the Project Plan/document checklist and specifications and should review all the certificates, approvals, trackers, logbooks etcetera and verify that nothing is missed out. Also, the PM shall ensure that the NCRs, Shortfalls notices by the Customer/ Consultant are addressed rectified and approval is obtained for the same.

b. PM shall ensure that all the invoices are submitted and paid except the retainage including the cost for Change Order items where applicable.

c. Turn over Operations and Maintenance Manuals, User Guides, As-built drawings, Warranties, and other documents to the Customer/ Concerned Parties as required by Contract Conditions. Demonstrations/ Operation Training/ Maintenance Training etcetera shall also be provided based on the requirements and contract conditions and shall obtain handover signoffs from Customer/ Consultant.

d. A final bill should be submitted within one working day after all the deliverables including handing over of closure documents, trainings, demonstrations etcetera are completed in which shall include any missed-out claims, disputed claims etcetera and along with that the request for release of retainage shall also be submitted.

e. Certification of Customer and/ or Consultant to the effect that all works, and deliverables have been completed as per specifications laid down in the contract and defects noted and notified to Utthunga has been liquidated to the satisfaction of Customer/ Consultant.

f. PM Shall ensure that all the trackers completely updated and are duly checked in to the repository and they are archived.

g. The Project Manager shall with the approval of Delivery/ Head and Customer, initiate the process to demobilize resources from the site. The demobilization activity shall be smooth and after informing all the concerned departments/ agencies and after considering the requirements like providing spares and for activities during Defect Liability Period (DLP) as per the project specifications; information regarding the demobilization of human resource shall be passed on to HR Department well in advance at least one week ahead.

h. On completion of handing over formalities and successful completion of Defect Liability Period (DLP), the contract shall be closed on completion of following formalities:

i. Payment reconciliations, submission, and verifications that reconciliation of payment toward statutory provisions like GST or any other dues etc.

j. Approval for extension of Completion period, with or without compensation, as agreed upon among the parties.

k. The requirements of Customers shall be appreciated well after closure also including post-sales support.

l. As per the project requirement, if post-sales order support is required by the customer in terms of maintenance, Project manager shall prepare a comprehensive plan in terms of deployment of resources, such as manpower and necessary equipment and tools. A separate monitoring sheet shall be prepared, a customer acknowledged timesheet shall be obtained, and information shall be passed on to accounts for the invoicing process.

m. The project manager shall monitor the maintenance activities and coordinate with customer as per plan for arranging resources on time and readiness of site as per the schedule agreed by customer.

n. Feedback shall be obtained from the Customer for assessment and for establishing steps to improve the quality of services further.

11 Information Security requirements

a. Procedures shall be implemented for the management of removable media in accordance with the classification scheme adopted by the organization.

b. Media containing information shall be protected against unauthorized access, misuse, or corruption during transportation.

c. The information security related requirements shall be included in the requirements for new information systems or enhancements to existing information systems.

d. The organization shall supervise and monitor the activity of outsourced system development.

e. Information security requirements for mitigating the risks associated with supplier's access to the organization's assets shall be agreed with the supplier and documented.

f. All relevant information security requirements shall be established and agreed with each supplier that may access, process, store, communicate, or provide IT infrastructure components for, the organization's information.

g. Agreements with suppliers shall include requirements to address the information security risks associated with information and communications technology services and product supply chain.

h. Organizations shall regularly monitor, review and audit supplier service delivery.

i. Changes to the provision of services by suppliers, including maintaining and improving existing information security policies, procedures, and controls, shall be managed, taking account of the criticality of business information, systems and processes involved and re-assessment of risks.

j. Confidentiality of data is maintained.

12 References

a. ISO 9001 Clauses 8.5,8.6,8.7

b. ISO 27001 Clauses 8.1, Controls A.6, A.7, A.8, A.15

c. ISMS Control Policies

d. Opportunity Management process UT/SM/OMP/SOP

e. Procurement process UT/SF/PRO/SOP

f. Inventory Management UT/SF/INVM/SOP

g. Customer Complaint and Escalation UT/PS/CCEP/SOP

13 Metrics

a. Ontime milestone completion

b. Project Completion (Schedule Variance)

c. Ontime Material receipt to site

d. Effort Variance

e. No.of quality nonconformance

f. Requirement stability index

g. Resource stability index

h. Number of Recovery Plans (revisions to project plan)

i. Budget VS Actual

j. CSAT

k.

14 Annexures

a. NIL

15 Forms

a. Document transmittal. UT/EPC/TMP/03

b. Daily work progress report UT/EPC/TMP/02

c. Nonconformance report UT/IM/MSP/TMP/16

d. Master document list UT/SF/PUR/TMP/23

e. Project Quality Plan UT/EPC/TMP/04

f. Quality assurance Plan UT/EPC/TMP/05

g. Completion certificate UT/EPC/TMP/01

h. Inspection Test Plan UT/EPC/TMP/09

i. Requirement Specifications-Projects UT/EPC/TMP/23

j. Project Plan Workbook - Projects UT/EPC/TMP/24

k. FAT Procedure Template – Projects UT/EPC/TMP/25

l. Commissioning Checklist Template – Projects UT/EPC/TMP/26